

Firehouse Capacity Analysis - EMS Optimization Model

Date: March 15, 2026
Scope: Capacity constraints in the MIP formulations allocating K EMS units across 48 Manhattan firehouses

1. Current Model Assumptions

1.1 Capacity Parameter in Code

The optimization configuration (`configs/optimization.yaml`) sets:

```
firehouse_capacity: 5
```

Every MIP formulation in `models.py` uses `capacity=5` as the default upper bound:

Model	Variable	Capacity Enforcement
Demand-Weighted (P2)	$x_i \in \{0, \dots, \text{capacity}\}$ (Integer)	Hard upper bound on <code>LpVariable</code>
P-Median (P2b)	$x_i \in \{0, 1\}$ (Binary)	Implicit cap = 1 (facility open/close)
Maximal Coverage (P2c)	$x_i \in \{0, \dots, \text{capacity}\}$ (Integer)	Hard upper bound on <code>LpVariable</code>
CBD-Focused	Same as above	Same

Baseline policies (`policies.py`) also accept `capacity=5` :

Policy	Capacity Enforcement
P0 - Uniform	Round-robin respects <code>capacity</code> ceiling
P1 - Demand-Proportional	Integer rounding capped at <code>capacity</code>
P0-Spatial (stratified)	Round-robin among selected firehouses, capped at <code>capacity</code>

Key finding: The capacity parameter exists and is enforced, but the default value of **5** is generous. Several models exploit it.

1.2 How the Constraint Appears in the MIP

```
# models.py – Demand-Weighted
x = pulp.LpVariable.dicts("x", firehouses,
    lowBound=0, upBound=capacity, cat="Integer")
```

The P-Median model uses **binary** x_i (open/close), so each opened firehouse effectively gets exactly 1 unit—capacity is not a binding concern there.

2. Current Allocation Patterns

2.1 K = 20 (Primary Scenario)

Policy	Max Units at 1 FH	# Active FHs (of 48)	Distribution
P0 (Uniform)	1	20	{0: 28, 1: 20}
P1 (Demand-Prop)	2	18	{0: 30, 1: 16, 2: 2}
P2 (Demand-Wt)	1	20	{0: 28, 1: 20}
P2b (P-Median)	1	20	{0: 28, 1: 20}
P2c (Max Coverage)	5	7	{0: 41, 1: 3, 2: 1, 5: 3}

At K=20, P2c is the only policy that hits the capacity ceiling, concentrating 5 units at 3 firehouses (15 of 20 units at 3 locations). All other policies stay at ≤ 2 units per firehouse.

2.2 Across All K Values

K	P0 max	P1 max	P2 max	P2b max	P2c max
20	1	2	1	1	5
30	1	3	5	1	5
40	1	4	5	1	5
48	1	5	5	1	5

- **P2c (Maximal Coverage)** always concentrates at the cap of 5, regardless of K.
- **P2 (Demand-Weighted)** starts concentrating once $K > 20$ (reaches cap at $K=30+$).
- **P1 (Demand-Proportional)** gradually concentrates as K grows.
- **P0 and P2b** never exceed 1 unit/firehouse (by design: uniform and binary p-median).

3. Realistic Capacity Recommendations

3.1 Research Findings

Source	Key Finding
FDNY structure	Each firehouse houses 1–3 fire companies (engine + ladder + battalion). Apparatus bays hold 1–3 vehicles typically.
FDNY EMS	Manhattan has only 6 dedicated EMS stations (not firehouses). FDNY EMS is a separate system from fire suppression with its own stations.
EMS-only stations	Purpose-built EMS stations can house 2–20 ambulances (Boston example: 11-bay for 20 ambulances). These are not standard firehouses.
Fire station design	Typical fire stations have 2–4 apparatus bays . EMS units sharing space must compete with engines, ladders, and special units.
NYC reality	FDNY firehouses already house engine + ladder companies. An additional EMS unit (ambulance + crew) requires a free bay, crew quarters, and supplies.

3.2 Recommended Capacity Constraints

Scenario	Recommended Cap	Rationale
Conservative / Realistic	2	Most Manhattan firehouses have 2–3 bays already occupied by engine + ladder companies. Fitting 1–2 EMS units is feasible with shared bays or apron staging.
Moderate	3	Allows for larger double-house stations (e.g., Battalion HQ stations) that may have extra bay space.
Current (generous)	5	Only achievable at purpose-built EMS stations, not typical firehouses. Unrealistic for most of the 48 Manhattan firehouses.

3.3 Recommendation

Use `firehouse_capacity: 2` as the default, with an optional override of 3 for identified large stations.

This ensures:

1. No firehouse is unrealistically over-loaded.
 2. The optimizer spreads units geographically (better spatial coverage).
 3. Results align with FDNY operational reality where EMS ambulances share space with fire companies.
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4. Impact Assessment

4.1 Would Capacity = 2 Change Optimal Solutions?

Policy	K=20 Impact	K=30 Impact	K=40+ Impact
P0 (Uniform)	None (already max=1)	None	None
P1 (Demand-Prop)	None (max=2)	Yes - forces spreading from 3 to 2	Yes - significant redistribution
P2 (Demand-Wt)	None (max=1 at K=20)	Yes - forces more firehouses open	Yes
P2b (P-Median)	None (binary)	None	None
P2c (Max Coverage)	YES - major change - must use 10+ FHs instead of 7	Yes	Yes

At K=20: Only P2c would be materially affected. Its objective value will decrease (less coverage or same coverage with worse concentration).

At K=30: P1, P2, and P2c are all affected. With cap=2, P1-demand uses 21 firehouses (vs 30 at cap=1) and achieves the best mean RT of 2.39 min. P2-optimised uses 25 firehouses at cap=2 with mean RT of 2.44 min. The optimizer is forced to open more firehouses, improving spatial dispersion.

At K=40+: Similar effects are amplified — see `capacity_sensitivity_analysis.md` for the full spectrum.

4.2 Is This a Problem?

No - it's a feature. The current P2c solution (3 firehouses with 5 units each at K=20) is operationally unrealistic. Forcing dispersion through capacity constraints produces solutions that could actually be implemented.

5. Code Snippets for Updating Capacity Constraints

5.1 Update `configs/optimization.yaml`

```
# Change from 5 to 2
firehouse_capacity: 2

# Or use per-firehouse capacities (future enhancement)
# firehouse_capacity_overrides:
# "Battalion 10/Engine 22/Ladder 13": 3 # large double-house
```

5.2 No Code Changes Needed in `models.py`

The capacity parameter is already properly implemented. Changing the YAML value from 5 to 2 propagates through `allocator.py` to all MIP formulations:

```
# In models.py – already correct:
x = pulp.LpVariable.dicts("x", firehouses,
    lowBound=0, upBound=capacity, cat="Integer")
```

5.3 Optional: Per-Firehouse Capacity (Future Enhancement)

To allow different capacities per firehouse (e.g., large stations get cap=3):

```
# In models.py – replace scalar capacity with dict
def build_demand_weighted(
    travel_time, demand, K,
    capacity=2, # default
    capacity_overrides: dict = None, # NEW
    solver_time_limit=300,
):
    firehouses = list(travel_time.index)
    precincts = [p for p in travel_time.columns if p in demand.index]

    # Per-firehouse capacity
    cap = {}
    for fh in firehouses:
        if capacity_overrides and fh in capacity_overrides:
            cap[fh] = capacity_overrides[fh]
        else:
            cap[fh] = capacity

    prob = pulp.LpProblem("DemandWeighted_EMS_Allocation", pulp.LpMinimize)

    # Use per-firehouse upper bound
    x = {fh: pulp.LpVariable(f"x_{fh}", lowBound=0, upBound=cap[fh], cat="Integer")}
    for fh in firehouses:
        # ... rest unchanged ...
```

6. Summary & Next Steps

Item	Status	Action
Capacity constraint exists in code	✓	No structural changes needed
Default capacity = 5 is too generous	⚠	Change to 2 in <code>optimization.yaml</code>
P2c solutions are unrealistic at all K	⚠	Will auto-fix with <code>cap=2</code>
P2 solutions unrealistic at $K \geq 30$	⚠	Will auto-fix with <code>cap=2</code>
P0, P2b unaffected	✓	No action needed
Per-firehouse capacity	➡ SOON	Nice-to-have for future phases

Recommended Immediate Action

1. Set `firehouse_capacity: 2` in `configs/optimization.yaml`
2. Re-run all optimization experiments at $K \in \{20, 30, 40, 48\}$
3. Compare new results with current results to quantify the impact on objective values
4. Update the spatially-stratified P0 analysis with the same capacity constraint

Appendix: Visualization

See `firehouse_capacity_analysis.png` for:

- Distribution of units per firehouse for each policy at $K=20$
- Maximum firehouse load trends across K values with realistic capacity thresholds
- Count of firehouses exceeding realistic capacity (>2 units)